

Use the following information for questions 1-4. Directed line segment $\overline{WZ}$.



1. How many equal parts is $\overline{WZ}$ partitioned into?
11
2. Make a mark where $\overline{WZ}$ is partitioned into a 2:3 ratio. Call this point X.
 $2 + 3 = 5$
 $2\left(\frac{11}{5}\right) = 4\frac{2}{5}$
3. Complete the statement.

Point X is $\frac{2}{5}$ of the distance from W to Z.

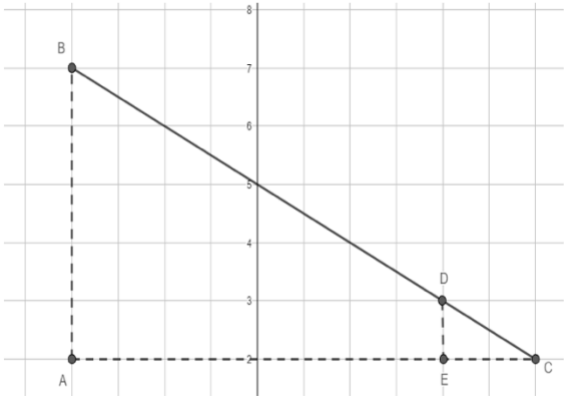
4. Complete the equation that can be used to find the distance of $\overline{WX}$.

$WX = \frac{2}{5} WZ$

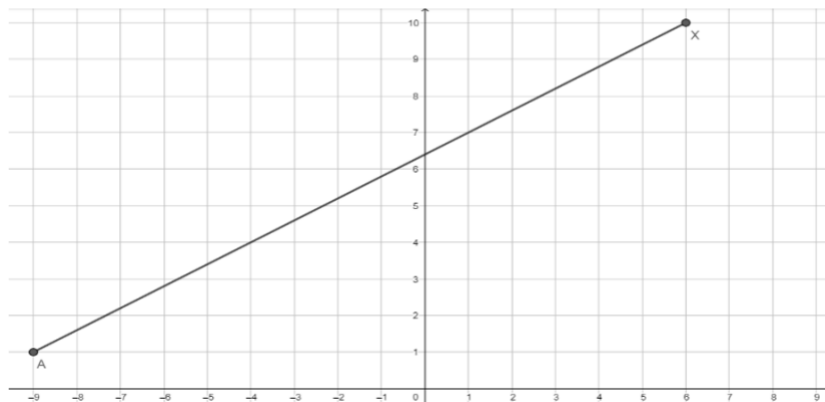
Point D partitions directed line segment $\overline{BC}$. Points $A(-4,2)$, $E(4,2)$, $D(4,3)$, $B(-4,7)$, and $C(6,2)$ are shown on the coordinate grid.

Use the distances to write the numeric ratio for each segment ratio. Then, choose the type of ratio the segment ratio represents.

Segment Ratio	Numeric Ratio	Type of Ratio
$CD:DB$	1:4	☒ part: part ☐ part: whole
$CE:CA$	1:5	☐ part: part ☒ part: whole
$DE:BA$	1:5	☐ part: part ☒ part: whole
$CD:CB$	1:5	☐ part: part ☒ part: whole



Use the following information for questions 5-7. Directed line segment AX is shown with endpoints at point $A(-9,1)$ and point $X(6,10)$.



Directed line segment AX is partitioned $\frac{4}{5}$ of the distance from A to X at point B .

5. What is the horizontal distance of $\overline{AB}$?

$$\frac{4}{5}(15) = 12$$

6. What is the vertical distance of $\overline{AB}$?

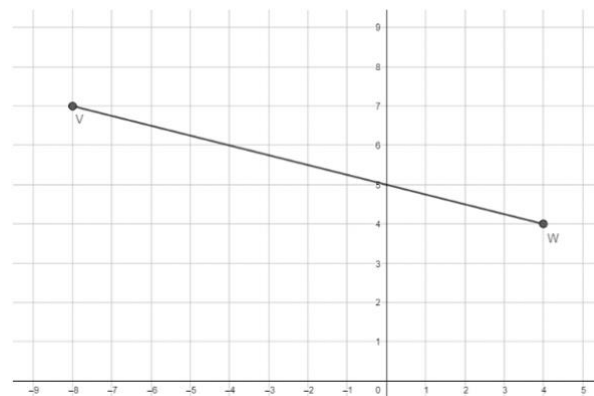
$$\frac{4}{5}(9) = \frac{36}{5} \text{ or } 7.2$$

7. What are the coordinates of point B ?

$$(-9 + 12, 1 + 7.2) = (3, 8.2)$$

Use the following information for questions 8-10.

Directed line segment VW has endpoints at $V(-8,7)$ and $W(4,4)$.



Directed line segment $\overline{VW}$ is partitioned $\frac{1}{4}$ of the distance from V to W at point R .

8. What is the horizontal distance of $\overline{VR}$?

$$\frac{1}{4}(4 - (-8)) = 3$$

9. What is the vertical distance of $\overline{VR}$?

$$\frac{1}{4}(7 - 4) = \frac{3}{4}$$

10. What are the coordinates of point R ?

$$(-8 + 3, 7 - \frac{3}{4}) = (-5, 6\frac{1}{4})$$